

Special Functions

S.1

Solution:

Using the definition of $\Gamma(z)$:

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$$

it can be shown that $\Gamma(z+1) = z\Gamma(z)$:

$$\begin{aligned} \Gamma(z+1) &= \int_0^{\infty} t^z e^{-t} dt = \underbrace{[-t^z e^{-t}]_0^{\infty}}_{=0} + \int_0^{\infty} z t^{z-1} e^{-t} dt = \\ &= z \int_0^{\infty} t^{z-1} e^{-t} dt = z\Gamma(z) \end{aligned}$$

We also have

$$\Gamma(1) = \int_0^{\infty} e^{-t} dt = [-e^{-t}]_0^{\infty} = 1$$

This gives

$$\Gamma(3) = \Gamma(2+1) = 2\Gamma(2) = 2\Gamma(1+1) = 2 \cdot 1 \cdot \Gamma(1) \rightarrow \Gamma(3) = 2$$

Next

$$\begin{aligned} \Gamma\left(\frac{1}{2}\right) &= \int_0^{\infty} t^{-\frac{1}{2}} e^{-t} dt \rightarrow [t = x^2 \rightarrow dt = 2x dx] = 2 \int_0^{\infty} x x^{-1} e^{-x^2} dx = \\ &= 2 \int_0^{\infty} e^{-x^2} dx = 2 \frac{\sqrt{\pi}}{2} \rightarrow \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \end{aligned}$$

Finally we will use the definition for $\Gamma(z)$, $\text{Re } z \leq 0$:

$$\Gamma(z) = \frac{\Gamma(z+n)}{(z+n-1)\dots(z+1)z}, \quad n \text{ chosen such that } \operatorname{Re}(z+n) > 0$$

This gives

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma\left(-\frac{1}{2}+1\right)}{-\frac{1}{2}} = \frac{\Gamma\left(\frac{1}{2}\right)}{-\frac{1}{2}} \rightarrow \Gamma\left(-\frac{1}{2}\right) = -2\sqrt{\pi}$$

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S.5

Solution:

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S.6

Solution:

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S.7

$$\sum_{-\infty}^{\infty} |J_n(r)|^2 = 1$$

Solution:

Using the Fourier expansion

$$e^{ir \sin \theta} = \sum_{-\infty}^{\infty} J_n(r) e^{in\theta}$$

and Parseval's Theorem

$$\begin{aligned}
\frac{1}{2\pi} \int_{-\pi}^{\pi} \overline{e^{ir \sin \theta}} e^{ir \sin \theta} d\theta &= \sum_{-\infty}^{\infty} \overline{J_n(r)} J_n(r) \overline{e^{in\theta}} e^{in\theta} \\
&\downarrow \\
\frac{1}{2\pi} \int_{-\pi}^{\pi} d\theta &= \sum_{-\infty}^{\infty} |J_n(r)|^2 \\
&\downarrow \\
\frac{1}{2\pi} 2\pi &= \sum_{-\infty}^{\infty} |J_n(r)|^2 = 1
\end{aligned}$$

□

S.11

$$\frac{d}{dx} \left(\frac{J_\nu(x)}{x^\nu} \right) = -\frac{J_{\nu+1}(x)}{x^\nu}$$

(b) Use (a) and

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

to calculate $J_{\frac{3}{2}}(x)$ (c) Show that $J'_0(x) = -J_1(x)$ **Solution:**

The power series is

$$\begin{aligned}
J_\nu(x) &= \left(\frac{x}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^k \\
&\downarrow \\
\frac{J_\nu(x)}{x^\nu} &= \left(\frac{1}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^k
\end{aligned}$$

This can be differentiated termwise as

$$\begin{aligned}
\frac{d}{dx} \left(\frac{J_\nu(x)}{x^\nu} \right) &= \left(\frac{1}{2} \right)^\nu \sum_{k=0}^{\infty} \frac{-2xk}{4k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4} \right)^{k-1} = \\
&= -\frac{x}{2} \left(\frac{1}{2} \right)^\nu \sum_{\substack{k=1 \\ *}}^{\infty} \frac{1}{(k-1)! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4} \right)^{k-1} = \\
&= -\frac{1}{x^\nu} \left(\frac{x}{2} \right)^{\nu+1} \sum_{k=1}^{\infty} \frac{1}{(k-1)! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4} \right)^{k-1} \rightarrow [k \rightarrow k+1] \rightarrow \\
&\rightarrow -\frac{1}{x^\nu} \left(\frac{x}{2} \right)^{\nu+1} \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + 1 + k + 1)} \left(-\frac{x^2}{4} \right)^k = \boxed{-\frac{J_{\nu+1}(x)}{x^\nu}}
\end{aligned}$$

* (The $k=0$ contribution is zero, since it is a constant and was differentiated.)

(b) Setting $\nu = \frac{1}{2}$ we get

$$\begin{aligned}
\frac{d}{dx} \left(\frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}} \right) &= -\frac{J_{\frac{3}{2}}(x)}{x^{\frac{1}{2}}} \\
\frac{d}{dx} \left(\frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}} \right) &= \frac{d}{dx} \left(\frac{\sqrt{\frac{2}{\pi x}} \sin x}{x^{\frac{1}{2}}} \right) = \frac{d}{dx} \left(\sqrt{\frac{2}{\pi}} \frac{\sin x}{x} \right) = \sqrt{\frac{2}{\pi}} \left(-\frac{1}{x^2} \sin x + \frac{1}{x} \cos x \right) \\
&\downarrow \\
J_{\frac{3}{2}}(x) &= x^{\frac{1}{2}} \sqrt{\frac{2}{\pi}} \left(\frac{\sin x}{x^2} - \frac{x}{x^2} \cos x \right) = \sqrt{\frac{2}{\pi}} x^{\frac{3}{2}} \left(\frac{\sin x - x \cos x}{x^3} \right)
\end{aligned}$$

(c) Using the result from (a)

$$\frac{d}{dx} \left(\frac{J_0(x)}{x^0} \right) = -\frac{J_1(x)}{x^0} \rightarrow \boxed{J_{0'}(x) = -J_1(x)}$$

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S.12

Solution:

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S.13

(a) $r^2 y'' + r y' = 0$

(b) $r^2 y'' + r y' - y = 0$

(c) $r^2 y'' + r y' + r^2 y = 0$

Solution:

Using a definition that the general solution to Bessel's differential equations

$$u'' + \frac{1}{r}u' + \left(\lambda - \frac{\nu^2}{r^2}\right)u = 0$$

are

$$\begin{aligned} & aJ_\nu(\sqrt{\lambda}r) + bY_\nu(\sqrt{\lambda}r), & \text{if } \lambda > 0 \\ & ar^\nu + br^{-\nu}, & \text{if } \lambda = 0, \nu \neq 0 \\ & a + b \ln r, & \text{if } \lambda = \nu = 0 \end{aligned}$$

(a)

$$r^2 y'' + r y' = 0 \rightarrow y'' + \frac{1}{r}y' = 0, \quad \lambda = \nu = 0$$

$$y(r) = a + b \ln r$$

(b)

$$r^2 y'' + r y' - y = 0 \rightarrow y'' + \frac{1}{r}y' - \frac{1}{r^2}y = 0, \quad \lambda = 0, \nu = 1$$

$$y(r) = ar + \frac{b}{r}$$

(c)

$$r^2 y'' + r y' + r^2 y = 0 \rightarrow y'' + \frac{1}{r}y' + y = 0, \quad \lambda = 1, \nu = 0$$

$$y(r) = aJ_0(r) + bY_0(r)$$

□

S.14

$$\mathcal{A}u = -\frac{1}{x}(xu')'$$

$$D_{\mathcal{A}} = \{u \in \mathcal{C}^2(I) \mid u \text{ bounded near } 0, u'(2) = 0\}$$

- (a) Determine a scalar product in which $\mathcal{A}$ is symmetric and positive semidefinite
- (b) Determine the eigenvalues and eigenvectors of $\mathcal{A}$.
- (c) Determine numerically the three lowest eigenvalues.
- (d) Let $u(x) = x$. Write down u 's expansion in eigenfunctions to $\mathcal{A}$. The coefficients should be given in integral form.

Solution:

(a)

$$(u \mid \mathcal{A}v) = \int_0^2 \overline{u(x)} \mathcal{A}v(x) w(x) dx$$

with $w(x) = x$

$$\begin{aligned} (u \mid \mathcal{A}v) &= - \int_0^2 \overline{u} (xv')' dx = - \underbrace{[\overline{u}(xv')]_0^2}_{=0} + \int_0^2 (\overline{u}'x) v' dx = \\ &= \underbrace{\int_0^2 (x\overline{u}') v' dx}_{= (*)} = \underbrace{[(x\overline{u}')v]_0^2}_{=0} - \int_0^2 (x\overline{u}')' v dx = (\mathcal{A}u \mid v) \end{aligned}$$

To show that it's positive semidefinite, start from (*) and set $u = v$:

$$\int_0^2 x \overline{u}' u' dx = \int_0^2 x |u'|^2 dx \geq 0$$

(b) Setting

$$\begin{aligned} \mathcal{A}u = \lambda u &\rightarrow -\frac{1}{x}(xu')' = -\frac{1}{x}(u' + xu'') = \lambda u \\ &\downarrow \\ -\frac{1}{x}xu'' - \frac{1}{x}u' - \lambda u &= 0 \rightarrow u'' + \frac{1}{x}u' + \lambda u = 0 \end{aligned}$$

which has the solutions (according to the definition of the Bessel's differential equation solutions):

$$\begin{aligned} aJ_0(\sqrt{\lambda}x) + bY_0(\sqrt{\lambda}x), & \quad \lambda > 0, \nu = 0 \\ a + b \ln r, & \quad \lambda = \nu = 0 \end{aligned}$$

Since Y_0 is unbounded near $x = 0$, b has to be zero to fulfill the requirement of u being bounded near zero.

$$u(x) = aJ_0(\sqrt{\lambda}x)$$

$$u'(2) = \sqrt{\lambda}aJ_0'(\sqrt{\lambda}2) = 0$$

This gives $\lambda_0 = 0$. The rest of the eigenvalues are given by

$$J_0'(2\sqrt{\lambda_k}) = 0 \rightarrow \lambda_k = \left(\frac{\beta_{0,k}}{2}\right)^2$$

The eigenfunctions are then given by

$$\varphi_{k(x)} = J_0\left(\frac{\beta_{0,k}}{2}x\right)$$

(c)

$$\lambda_0 = 0$$

$$\lambda_1 = \left(\frac{3.8317}{2}\right)^2 = 3.67$$

$$\lambda_2 = \left(\frac{7.0156}{2}\right)^2 = 12.31$$

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S.15

Solution:

□

S.16

Solution:

□

S.18**Solution:**☐

S.20**Solution:**☐